CNDC PROJECT REPORT

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1. Executive Summary

This report presents the complete network design and IP configuration for the CNDC (Computer Networks and Data Communications) project. The project involves designing and implementing a multi-campus enterprise network connecting six campuses: Campus 99 (main hub), Campus 100, Campus 154, Campus 153, Campus 79, and Media Villa.

The network employs a hierarchical design using VLAN segmentation, inter-VLAN routing, DHCP services, static IP allocation for critical infrastructure, and a centralized WAN connection to the ISP through EDGE-99.

2. Project Scope

The scope of this project encompasses the full design and simulation of a multi-campus enterprise network. The key deliverables include:

2.1 Network Coverage

- Design and configuration of 6 campuses / sites
- Total VLANs configured: 30+ across all campuses
- End devices: PCs, Printers, FTP Servers, Web/Email/CMS/ZABDESK Servers
- Network simulation performed in Cisco Packet Tracer

2.2 Technical Scope

- Sub-interface configuration on all campus routers for Inter-VLAN routing
- DHCP pool configuration for dynamic IP assignment per VLAN
- Static IP assignment for all servers, printers, and core switches
- WAN connectivity: EDGE-99 acts as the central hub connecting all campuses
- ISP connection via Serial link (11.11.11.0/30) for internet access
- Routing between campuses using static/dynamic routing via EDGE-99

2.3 Out of Scope

- Physical hardware procurement and installation
- Security policies, firewall configuration, and ACLs (beyond basic design)
- Cloud integration or wireless (Wi-Fi) infrastructure
- Real-world deployment and testing

3. Network Topology Overview

The network follows a hub-and-spoke topology with EDGE-99 (Campus 99) acting as the central router connecting all remote campuses. The topology diagram (Cisco Packet Tracer simulation) illustrates the full inter-campus connectivity.

Key topology components:

- EDGE-99: Central edge router — connects to ISP, R-100, R-154, R-153, and R-MEDIA
- R-100: Router for Campus 100 (13 VLANs — Computer Labs)
- R-154: Router for Campus 154 (5 VLANs — Academic departments)
- R-153: Router for Campus 153 (3 VLANs — Classrooms, Admission, Academic)
- R-79: Router for Campus 79 (6 VLANs — Administrative departments)
- R-MEDIA: Router for Media Villa (3 VLANs — Classrooms, Faculty, Academic)

3.1 Inter-Campus WAN Links

Link	Interface	IP Address	Network	Medium
EDGE-99 ↔ ISP	Se0/1/0	11.11.11.1	11.11.11.0/30	WAN/ISP
EDGE-99 ↔ R-100	Gi0/1	23.12.145.109/110	23.12.145.108/30	Ethernet
EDGE-99 ↔ R-154	Gi0/2	23.12.145.113/114	23.12.145.112/30	Ethernet
EDGE-99 ↔ R-153	Se0/0/0	23.12.145.117/118	23.12.145.116/30	Serial
EDGE-99 ↔ R-MEDIA	Se0/1/1	23.12.145.125/126	23.12.145.124/30	Serial
EDGE-99 ↔ R-79	Se0/0/1	23.12.145.121/122	23.12.145.120/30	Serial

4. Campus 99 — Main Campus (EDGE-99)

Campus 99 is the primary campus hosting the EDGE-99 router, which serves as the network hub. It houses the central Data Center with critical servers and connects to the ISP for internet access.

4.1 VLAN Configuration & IP Allocation

VLAN	Name	Network	Gateway	DHCP Range	Type
140	CLASSROOMS-99	23.12.145.0/29	23.12.145.1	23.12.145.2 – .6	DHCP
150	FACULTY-99	23.12.145.48/28	23.12.145.49	23.12.145.50 – .62	DHCP
160	ACADEMIC-99	23.12.145.16/30	23.12.145.17	23.12.145.18	DHCP
170	IT-99	23.12.145.20/30	23.12.145.21	23.12.145.22	DHCP
180	EXAMINATION	23.12.145.24/29	23.12.145.25	23.12.145.26 – .30	DHCP
190	ADMIN-99	23.12.145.32/29	23.12.145.33	23.12.145.35 – .38	DHCP
200	DATACENTER	23.12.145.40/29	23.12.145.41	23.12.145.42 – .46	Static

4.2 EDGE-99 Interface Configuration

Interface	VLAN	Department	IP Address	Subnet Mask
Gi0/0.140	140	CLASSROOMS-99	23.12.145.1	255.255.255.248
Gi0/0.150	150	FACULTY-99	23.12.145.49	255.255.255.240
Gi0/0.160	160	ACADEMIC-99	23.12.145.17	255.255.255.252
Gi0/0.170	170	IT-99	23.12.145.21	255.255.255.252
Gi0/0.180	180	EXAMINATION	23.12.145.25	255.255.255.248
Gi0/0.190	190	ADMIN-99	23.12.145.33	255.255.255.248
Gi0/0.200	200	DATACENTER	23.12.145.41	255.255.255.248
Gi0/1	—	Link to R-100	23.12.145.109	255.255.255.252
Se0/1/0	—	ISP Link (WAN)	11.11.11.1	255.255.255.252
Gi0/2	—	Link to R-154	23.12.145.113	255.255.255.252
Se0/0/0	—	Link to R-153	23.12.145.117	255.255.255.252
Se0/0/1	—	Link to R-79	23.12.145.121	255.255.255.252
Se0/1/1	—	Link to R-MEDIA	23.12.145.125	255.255.255.252

4.3 Data Center & Static Devices

Device	VLAN	IP Address	Subnet Mask	Gateway	Type
Web-Server	200	23.12.145.42	255.255.255.248	23.12.145.41	Static
ZABDESK-Server	200	23.12.145.43	255.255.255.248	23.12.145.41	Static
Email-Server	200	23.12.145.44	255.255.255.248	23.12.145.41	Static
CMS-Server	200	23.12.145.45	255.255.255.248	23.12.145.41	Static
Printer-Exam	180	23.12.145.27	255.255.255.248	23.12.145.25	Static
Printer-Faculty	150	23.12.145.50	255.255.255.240	23.12.145.49	Static

4.4 Testing

Same VLAN Ping

Web Server -> ZABDESK Server

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 23.12.145.43

Pinging 23.12.145.43 with 32 bytes of data:

Reply from 23.12.145.43: bytes=32 time<1ms TTL=128
Reply from 23.12.145.43: bytes=32 time<1ms TTL=128
Reply from 23.12.145.43: bytes=32 time<1ms TTL=128
Reply from 23.12.145.43: bytes=32 time=1ms TTL=128

Ping statistics for 23.12.145.43:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Inter VLAN Ping

Web Server -> Classroom gateway & Admin gateway

```
C:\>ping 23.12.145.1

Pinging 23.12.145.1 with 32 bytes of data:

Reply from 23.12.145.1: bytes=32 time<1ms TTL=255
Reply from 23.12.145.1: bytes=32 time<1ms TTL=255
Reply from 23.12.145.1: bytes=32 time=8ms TTL=255
Reply from 23.12.145.1: bytes=32 time<1ms TTL=255

Ping statistics for 23.12.145.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 8ms, Average = 2ms

C:\>ping 23.12.145.33

Pinging 23.12.145.33 with 32 bytes of data:

Reply from 23.12.145.33: bytes=32 time<1ms TTL=255
Reply from 23.12.145.33: bytes=32 time=2ms TTL=255
Reply from 23.12.145.33: bytes=32 time<1ms TTL=255
Reply from 23.12.145.33: bytes=32 time<1ms TTL=255

Ping statistics for 23.12.145.33:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

Cross Campus Ping

Web Server -> Campus 100 PC

```
C:\>ping 23.12.151.2

Pinging 23.12.151.2 with 32 bytes of data:

Request timed out.
Reply from 23.12.151.2: bytes=32 time=1ms TTL=126
Reply from 23.12.151.2: bytes=32 time=10ms TTL=126
Reply from 23.12.151.2: bytes=32 time=11ms TTL=126

Ping statistics for 23.12.151.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 11ms, Average = 7ms
```

Web Server -> Campus 154 PC

```
C:\>ping 23.12.145.130

Pinging 23.12.145.130 with 32 bytes of data:

Request timed out.
Reply from 23.12.145.130: bytes=32 time=10ms TTL=126
Reply from 23.12.145.130: bytes=32 time<1ms TTL=126
Reply from 23.12.145.130: bytes=32 time<1ms TTL=126

Ping statistics for 23.12.145.130:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 3ms
```

Web Server -> Campus 153 PC

```
C:\>ping 192.168.3.18

Pinging 192.168.3.18 with 32 bytes of data:

Reply from 192.168.3.18: bytes=32 time=23ms TTL=126
Reply from 192.168.3.18: bytes=32 time=1ms TTL=126
Reply from 192.168.3.18: bytes=32 time=1ms TTL=126
Reply from 192.168.3.18: bytes=32 time=15ms TTL=126

Ping statistics for 192.168.3.18:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 23ms, Average = 10ms
```

Web Server -> Campus 79 PC

```
C:\>ping 192.168.4.2

Pinging 192.168.4.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.4.2: bytes=32 time=10ms TTL=126
Reply from 192.168.4.2: bytes=32 time=2ms TTL=126
Reply from 192.168.4.2: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.4.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 10ms, Average = 4ms
```

Web Server -> Media Villa PC

```
C:\>ping 192.168.5.2

Pinging 192.168.5.2 with 32 bytes of data:

Reply from 192.168.5.2: bytes=32 time=2ms TTL=126
Reply from 192.168.5.2: bytes=32 time=10ms TTL=126
Reply from 192.168.5.2: bytes=32 time=10ms TTL=126
Reply from 192.168.5.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.5.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 10ms, Average = 5ms

C:\>
```

ISP Ping

```
C:\>ping 11.11.11.2

Pinging 11.11.11.2 with 32 bytes of data:

Reply from 11.11.11.2: bytes=32 time=2ms TTL=253
Reply from 11.11.11.2: bytes=32 time=2ms TTL=253
Reply from 11.11.11.2: bytes=32 time=2ms TTL=253
Reply from 11.11.11.2: bytes=32 time=2ms TTL=253

Ping statistics for 11.11.11.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 2ms, Average = 2ms
```

Internet Ping

```
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=1ms TTL=253
Reply from 8.8.8.8: bytes=32 time=1ms TTL=253
Reply from 8.8.8.8: bytes=32 time=1ms TTL=253
Reply from 8.8.8.8: bytes=32 time=1ms TTL=253

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

EDGE-99 Router:

```
EDGE-99#sh running-config
Building configuration...

Current configuration : 6620 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname EDGE-99
!
!
!
!
ip dhcp excluded-address 23.12.145.37 23.12.145.42
ip dhcp excluded-address 23.12.145.9
ip dhcp excluded-address 23.12.145.25
ip dhcp excluded-address 23.12.145.41 23.12.145.46
ip dhcp excluded-address 23.12.145.13
ip dhcp excluded-address 23.12.145.27
ip dhcp excluded-address 23.12.145.10
ip dhcp excluded-address 23.12.145.49
ip dhcp excluded-address 23.12.145.50
ip dhcp excluded-address 192.168.1.193
 --More--
```

```
11:00:46: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.2.185 on GigabitEthernet0/1 from LOADING to FULL,
Loading Done
ip dhcp excluded-address 192.168.1.254
 --More--
11:00:46: %OSPF-5-ADJCHG: Process 1, Nbr 23.12.145.161 on GigabitEthernet0/2 from LOADING to FULL,
Loading Done
ip dhcp excluded-address 192.168.2.97
ip dhcp excluded-address 192.168.2.126
ip dhcp excluded-address 192.168.2.185
ip dhcp excluded-address 192.168.2.190
ip dhcp excluded-address 23.12.151.61
ip dhcp excluded-address 23.12.151.189
ip dhcp excluded-address 23.12.151.190
ip dhcp excluded-address 192.168.1.61
ip dhcp excluded-address 192.168.1.62
ip dhcp excluded-address 23.12.145.1
ip dhcp excluded-address 23.12.145.17
ip dhcp excluded-address 23.12.145.21
ip dhcp excluded-address 23.12.145.33
ip dhcp excluded-address 23.12.145.34
ip dhcp excluded-address 23.12.145.41
!
ip dhcp pool VLAN140-CLASS99
 network 23.12.145.0 255.255.255.248
 default-router 23.12.145.1
 dns-server 8.8.8.8
ip dhcp pool VLAN160-ACAD99
 network 23.12.145.16 255.255.255.252
```

```
 default-router 23.12.145.17
 dns-server 8.8.8.8
ip dhcp pool VLAN170-IT99
 network 23.12.145.20 255.255.255.252
 default-router 23.12.145.21
 dns-server 8.8.8.8
ip dhcp pool VLAN180-EXAM
 network 23.12.145.24 255.255.255.248
 default-router 23.12.145.25
 dns-server 8.8.8.8
ip dhcp pool VLAN190-ADMIN99
 network 23.12.145.32 255.255.255.248
 default-router 23.12.145.33
 dns-server 8.8.8.8
ip dhcp pool VLAN200-DATACENTER
 network 23.12.145.40 255.255.255.248
 default-router 23.12.145.41
 dns-server 8.8.8.8
ip dhcp pool VLAN150-FAC99
 network 23.12.145.48 255.255.255.240
 default-router 23.12.145.49
 dns-server 8.8.8.8
!
```

```
interface GigabitEthernet0/0
 no ip address
 ip nat inside
 duplex auto
 speed auto
!
interface GigabitEthernet0/0.140
 encapsulation dot1Q 140
 ip address 23.12.145.1 255.255.255.248
 ip nat inside
!
interface GigabitEthernet0/0.150
 encapsulation dot1Q 150
 ip address 23.12.145.49 255.255.255.240
 ip nat inside
!
interface GigabitEthernet0/0.160
 encapsulation dot1Q 160
 ip address 23.12.145.17 255.255.255.252
 ip nat inside
!
interface GigabitEthernet0/0.170
 encapsulation dot1Q 170
 ip address 23.12.145.21 255.255.255.252
 ip nat inside
```

```
interface GigabitEthernet0/0.180
 encapsulation dot1Q 180
 ip address 23.12.145.25 255.255.255.248
 ip nat inside
!
interface GigabitEthernet0/0.190
 encapsulation dot1Q 190
 ip address 23.12.145.33 255.255.255.248
 ip nat inside
!
interface GigabitEthernet0/0.200
 encapsulation dot1Q 200
 ip address 23.12.145.41 255.255.255.248
 ip nat inside
 ip access-group SZABIST-ACL out
!
interface GigabitEthernet0/1
 ip address 23.12.145.109 255.255.255.252
 ip nat inside
 duplex auto
 speed auto
!
interface GigabitEthernet0/2
 ip address 23.12.145.113 255.255.255.252
 ip nat inside
 duplex auto
 speed auto
!
```

```
interface Serial0/0/0
 ip address 23.12.145.117 255.255.255.252
 ip nat inside
!
interface Serial0/0/1
 ip address 23.12.145.121 255.255.255.252
 ip nat inside
 clock rate 2000000
!
interface Serial0/1/0
 ip address 11.11.11.1 255.255.255.252
 ip nat outside
!
interface Serial0/1/1
 ip address 23.12.145.125 255.255.255.252
 ip nat inside
 clock rate 64000
!
interface Serial0/2/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/2/1
 no ip address
 clock rate 2000000
 shutdown
```

```
interface Serial0/3/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/3/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
router ospf 1
 log-adjacency-changes
 network 23.12.145.0 0.0.0.255 area 0
 network 23.12.151.0 0.0.0.255 area 0
 network 10.0.0.0 0.0.0.255 area 0
 network 192.168.1.0 0.0.0.255 area 0
 network 192.168.2.0 0.0.0.255 area 0
 network 11.11.11.0 0.0.0.3 area 0
 network 23.12.145.108 0.0.0.3 area 0
 network 23.12.145.112 0.0.0.3 area 0
 network 23.12.145.120 0.0.0.3 area 0
 network 23.12.145.124 0.0.0.3 area 0
 default-information originate
!
```

```
ip nat inside source list 1 interface Serial0/1/0 overload
ip classless
ip route 0.0.0.0 0.0.0.0 11.11.11.2
!
ip flow-export version 9
!
!
access-list 1 permit 23.12.145.0 0.0.0.255
access-list 1 permit 23.12.151.0 0.0.0.255
access-list 1 permit 192.168.1.0 0.0.0.255
access-list 1 permit 192.168.2.0 0.0.0.255
access-list 1 permit 192.168.3.0 0.0.0.255
access-list 1 permit 192.168.4.0 0.0.0.255
access-list 1 permit 192.168.5.0 0.0.0.255
ip access-list extended SZABIST-ACL
 permit ip 23.12.145.48 0.0.0.15 host 23.12.145.50
 permit ip 23.12.145.64 0.0.0.31 host 23.12.145.50
 permit ip 192.168.2.96 0.0.0.31 host 23.12.145.50
 permit ip 192.168.5.16 0.0.0.15 host 23.12.145.50
 deny ip any host 23.12.145.50
 permit ip 23.12.145.48 0.0.0.15 host 192.168.2.126
 permit ip 23.12.145.64 0.0.0.31 host 192.168.2.126
 permit ip 192.168.2.96 0.0.0.31 host 192.168.2.126
 permit ip 192.168.5.16 0.0.0.15 host 192.168.2.126
 deny ip any host 192.168.2.126
 permit ip 23.12.145.48 0.0.0.15 host 23.12.145.93
 permit ip 23.12.145.64 0.0.0.31 host 23.12.145.93
 permit ip 192.168.2.96 0.0.0.31 host 23.12.145.93
```

```
permit ip 192.168.5.16 0.0.0.15 host 23.12.145.93
deny ip any host 23.12.145.93
permit ip 23.12.145.48 0.0.0.15 host 192.168.5.27
permit ip 23.12.145.64 0.0.0.31 host 192.168.5.27
permit ip 192.168.2.96 0.0.0.31 host 192.168.5.27
permit ip 192.168.5.16 0.0.0.15 host 192.168.5.27
deny ip any host 192.168.5.27
permit ip 23.12.145.48 0.0.0.15 host 23.12.145.43
permit ip 23.12.145.64 0.0.0.31 host 23.12.145.43
permit ip 192.168.2.96 0.0.0.31 host 23.12.145.43
permit ip 192.168.5.16 0.0.0.15 host 23.12.145.43
permit ip 23.12.151.0 0.0.0.63 host 23.12.145.43
permit ip 23.12.151.64 0.0.0.63 host 23.12.145.43
permit ip 23.12.151.128 0.0.0.63 host 23.12.145.43
permit ip 23.12.151.192 0.0.0.63 host 23.12.145.43
permit ip 192.168.1.0 0.0.0.63 host 23.12.145.43
permit ip 192.168.1.64 0.0.0.63 host 23.12.145.43
permit ip 192.168.1.192 0.0.0.63 host 23.12.145.43
permit ip 192.168.2.64 0.0.0.15 host 23.12.145.43
deny ip any host 23.12.145.43
permit ip any any
!
```

SW-99-CORE:

Physical Config CLI Attributes [redacted]

```
!
!
interface GigabitEthernet1/0/1
 switchport mode trunk
!
interface GigabitEthernet1/0/2
 switchport mode trunk
!
interface GigabitEthernet1/0/3
 switchport mode trunk
!
interface GigabitEthernet1/0/4
 switchport mode trunk
!
interface GigabitEthernet1/0/5
 switchport mode trunk
!
interface GigabitEthernet1/0/6
!
interface GigabitEthernet1/0/7
!
interface GigabitEthernet1/0/8
!
interface GigabitEthernet1/0/9
!
interface GigabitEthernet1/0/10
!
interface GigabitEthernet1/0/11
 --More--
```

```
!
interface GigabitEthernet1/0/12
!
interface GigabitEthernet1/0/13
!
interface GigabitEthernet1/0/14
!
interface GigabitEthernet1/0/15
!
interface GigabitEthernet1/0/16
!
interface GigabitEthernet1/0/17
!
interface GigabitEthernet1/0/18
!
interface GigabitEthernet1/0/19
!
interface GigabitEthernet1/0/20
!
interface GigabitEthernet1/0/21
!
interface GigabitEthernet1/0/22
!
interface GigabitEthernet1/0/23
!
interface GigabitEthernet1/0/24
!
interface GigabitEthernet1/1/1
```

```
interface GigabitEthernet1/1/2
!
interface GigabitEthernet1/1/3
!
interface GigabitEthernet1/1/4
!
interface Vlan1
 no ip address
 shutdown
!
interface Vlan190
 mac-address 0000.0c89.1001
 ip address 23.12.145.34 255.255.255.248
!
ip default-gateway 23.12.145.33
ip classless
!
ip flow-export version 9
```

5. Campus 100 — Computer Labs

Campus 100 is connected to EDGE-99 via a Gigabit Ethernet link. It hosts 13 VLANs including specialized computer labs (CS, AI, Gaming, Smart Lab) and administrative departments. Router R-100 performs inter-VLAN routing.

5.1 VLAN & DHCP Configuration

VLAN	Name	Network	Gateway	DHCP Range	Type
10	CS-LAB	23.12.151.0/26	23.12.151.1	.2–.60	DHCP
20	AI-LAB	23.12.151.64/26	23.12.151.65	.66–.125	DHCP
30	LAB3	23.12.151.128/26	23.12.151.129	.130–.188	DHCP
40	LAB4	23.12.151.192/26	23.12.151.193	.194–.252	DHCP
50	LAB5	192.168.1.0/26	192.168.1.1	.2–.60	DHCP
60	LAB6	192.168.1.64/26	192.168.1.65	.66–.125	DHCP
70	SMART-LAB	192.168.1.192/26	192.168.1.193	.194–.253	DHCP
80	GAMING-LAB	192.168.2.64/28	192.168.2.65	.66–.77	DHCP
90	CLASSROOMS	192.168.2.128/28	192.168.2.129	.130–.142	DHCP
100	FACULTY	192.168.2.96/27	192.168.2.97	.98–.126	DHCP
110	ACADEMIC	192.168.2.144/28	192.168.2.145	.146–.158	DHCP
120	ADMIN	192.168.2.160/29	192.168.2.161	.162–.166	DHCP
130	IT-DEPT	192.168.2.184/29	192.168.2.185	.186–.190	DHCP

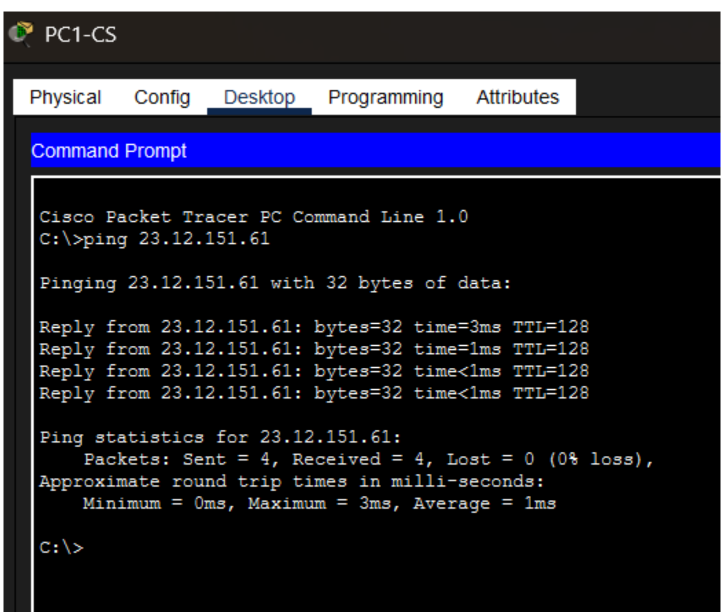
5.2 Static IPs — Servers, Printers & Core Switch

Device	IP Address	Subnet Mask	Gateway	Type
FTP-Server-CS	23.12.151.61	255.255.255.192	23.12.151.1	Static
Printer-CS	23.12.151.62	255.255.255.192	23.12.151.1	Static
Printer-AI	23.12.151.126	255.255.255.192	23.12.151.65	Static
FTP-Server-L3	23.12.151.189	255.255.255.192	23.12.151.129	Static
Printer-L3	23.12.151.190	255.255.255.192	23.12.151.129	Static
Printer-L4	23.12.151.253	255.255.255.192	23.12.151.193	Static
FTP-Server-L5	192.168.1.61	255.255.255.192	192.168.1.1	Static
Printer-L5	192.168.1.62	255.255.255.192	192.168.1.1	Static
Printer-SM	192.168.1.254	255.255.255.192	192.168.1.193	Static
Printer-GM	192.168.2.78	255.255.255.240	192.168.2.65	Static
Printer-FAC	192.168.2.126	255.255.255.224	192.168.2.97	Static
Printer-IT	192.168.2.190	255.255.255.248	192.168.2.185	Static
SW-100-CORE	192.168.2.163	255.255.255.248	192.168.2.161	Static

SW-100-CORE Management IP: 192.168.2.163 (VLAN 120, Gateway: 192.168.2.161)

5.3 Testing

Same VLAN Ping



Inter-VLAN Ping

```
C:\>ping 192.168.2.97

Pinging 192.168.2.97 with 32 bytes of data:

Reply from 192.168.2.97: bytes=32 time=10ms TTL=255
Reply from 192.168.2.97: bytes=32 time<1ms TTL=255
Reply from 192.168.2.97: bytes=32 time<1ms TTL=255
Reply from 192.168.2.97: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.2.97:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms
```

Cross-Campus Ping

```
C:\>ping 23.12.145.130

Pinging 23.12.145.130 with 32 bytes of data:

Reply from 23.12.145.130: bytes=32 time<1ms TTL=125
Reply from 23.12.145.130: bytes=32 time=9ms TTL=125
Reply from 23.12.145.130: bytes=32 time<1ms TTL=125
Reply from 23.12.145.130: bytes=32 time=20ms TTL=125

Ping statistics for 23.12.145.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 20ms, Average = 7ms
```

ISP Ping

```
C:\>ping 11.11.11.2

Pinging 11.11.11.2 with 32 bytes of data:

Reply from 11.11.11.2: bytes=32 time=4ms TTL=253
Reply from 11.11.11.2: bytes=32 time=4ms TTL=253
Reply from 11.11.11.2: bytes=32 time=4ms TTL=253
Reply from 11.11.11.2: bytes=32 time=4ms TTL=253

Ping statistics for 11.11.11.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 4ms, Average = 4ms

C:\>
```

Internet Ping

```
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=4ms TTL=253
Reply from 8.8.8.8: bytes=32 time=3ms TTL=253
Reply from 8.8.8.8: bytes=32 time=3ms TTL=253
Reply from 8.8.8.8: bytes=32 time=2ms TTL=253

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 4ms, Average = 3ms
```

R-100 Router:

```
!
ip dhcp excluded-address 23.12.151.1
ip dhcp excluded-address 23.12.151.65
ip dhcp excluded-address 23.12.151.129
ip dhcp excluded-address 23.12.151.193
ip dhcp excluded-address 192.168.1.1
ip dhcp excluded-address 192.168.1.65
ip dhcp excluded-address 192.168.1.193
ip dhcp excluded-address 192.168.2.65
ip dhcp excluded-address 192.168.2.97
ip dhcp excluded-address 192.168.2.113
ip dhcp excluded-address 192.168.2.145
ip dhcp excluded-address 192.168.2.161
ip dhcp excluded-address 192.168.2.185
ip dhcp excluded-address 23.12.151.61
ip dhcp excluded-address 23.12.151.62
ip dhcp excluded-address 23.12.151.126
ip dhcp excluded-address 23.12.151.189
ip dhcp excluded-address 23.12.151.190
ip dhcp excluded-address 23.12.151.253
ip dhcp excluded-address 192.168.1.61
ip dhcp excluded-address 192.168.1.62
ip dhcp excluded-address 192.168.1.254
ip dhcp excluded-address 192.168.2.78
ip dhcp excluded-address 192.168.2.126
ip dhcp excluded-address 192.168.2.190
ip dhcp excluded-address 192.168.2.129
```

```
ip dhcp pool VLAN10-CS
 network 23.12.151.0 255.255.255.192
 default-router 23.12.151.1
 dns-server 8.8.8.8
ip dhcp pool VLAN20-AI
 network 23.12.151.64 255.255.255.192
 default-router 23.12.151.65
 dns-server 8.8.8.8
ip dhcp pool VLAN30-LAB3
 network 23.12.151.128 255.255.255.192
 default-router 23.12.151.129
 dns-server 8.8.8.8
ip dhcp pool VLAN40-LAB4
 network 23.12.151.192 255.255.255.192
 default-router 23.12.151.193
 dns-server 8.8.8.8
ip dhcp pool VLAN50-LAB5
 network 192.168.1.0 255.255.255.192
 default-router 192.168.1.1
 dns-server 8.8.8.8
ip dhcp pool VLAN60-LAB6
 network 192.168.1.64 255.255.255.192
 default-router 192.168.1.65
 dns-server 8.8.8.8
ip dhcp pool VLAN70-SMART
 network 192.168.1.192 255.255.255.192
 default-router 192.168.1.193
 dns-server 8.8.8.8
```

```
ip dhcp pool VLAN80-GAME
 network 192.168.2.64 255.255.255.240
 default-router 192.168.2.65
 dns-server 8.8.8.8
ip dhcp pool VLAN110-ACAD
 network 192.168.2.144 255.255.255.240
 default-router 192.168.2.145
 dns-server 8.8.8.8
ip dhcp pool VLAN120-ADMIN
 network 192.168.2.160 255.255.255.248
 default-router 192.168.2.161
 dns-server 8.8.8.8
ip dhcp pool VLAN130-IT
 network 192.168.2.184 255.255.255.248
 default-router 192.168.2.185
 dns-server 8.8.8.8
ip dhcp pool VLAN100-FAC
 network 192.168.2.96 255.255.255.224
 default-router 192.168.2.97
 dns-server 8.8.8.8
ip dhcp pool VLAN90-CLASS
 network 192.168.2.128 255.255.255.240
 default-router 192.168.2.129
 dns-server 8.8.8.8
!
!
```

```
!
interface GigabitEthernet0/0
 ip address 23.12.145.110 255.255.255.252
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/1.10
 encapsulation dot1Q 10
 ip address 23.12.151.1 255.255.255.192
!
interface GigabitEthernet0/1.20
 encapsulation dot1Q 20
 ip address 23.12.151.65 255.255.255.192
!
interface GigabitEthernet0/1.30
 encapsulation dot1Q 30
 ip address 23.12.151.129 255.255.255.192
!
interface GigabitEthernet0/1.40
 encapsulation dot1Q 40
 ip address 23.12.151.193 255.255.255.192
!
```

```
interface GigabitEthernet0/1.50
 encapsulation dot1Q 50
 ip address 192.168.1.1 255.255.255.192
!
interface GigabitEthernet0/1.60
 encapsulation dot1Q 60
 ip address 192.168.1.65 255.255.255.192
!
interface GigabitEthernet0/1.70
 encapsulation dot1Q 70
 ip address 192.168.1.193 255.255.255.192
!
interface GigabitEthernet0/1.80
 encapsulation dot1Q 80
 ip address 192.168.2.65 255.255.255.240
!
interface GigabitEthernet0/1.90
 encapsulation dot1Q 90
 ip address 192.168.2.129 255.255.255.240
!
interface GigabitEthernet0/1.100
 encapsulation dot1Q 100
 ip address 192.168.2.97 255.255.255.224
!
interface GigabitEthernet0/1.110
 encapsulation dot1Q 110
 ip address 192.168.2.145 255.255.255.240
!
```

```
interface GigabitEthernet0/1.120
 encapsulation dot1Q 120
 ip address 192.168.2.161 255.255.255.248
!
interface GigabitEthernet0/1.130
 encapsulation dot1Q 130
 ip address 192.168.2.185 255.255.255.248
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
router ospf 1
 log-adjacency-changes
 network 23.12.145.108 0.0.0.3 area 0
 network 23.12.151.0 0.0.0.255 area 0
 network 192.168.1.0 0.0.0.255 area 0
 network 192.168.2.0 0.0.0.255 area 0
!
ip classless
ip route 0.0.0.0 0.0.0.0 23.12.145.109
```

6. Campus 154

Campus 154 connects to EDGE-99 via a Gigabit Ethernet link. It hosts 5 VLANs for academic and administrative departments including a Mechatronics department.

6.1 R-154 Sub-Interfaces & VLANs

VLAN	Name	Network	Gateway	DHCP Range	Type
210	CLASSROOMS-154	23.12.145.128/28	23.12.145.129	130–.142	DHCP
220	FACULTY-154	23.12.145.64/27	23.12.145.65	.66–.94	DHCP
230	ACADEMIC-154	23.12.145.144/29	23.12.145.145	146–.150	DHCP
240	ADMIN-154	23.12.145.152/29	23.12.145.153	154–.158	DHCP
250	MECHATRONICS-154	23.12.145.160/27	23.12.145.161	162–.190	DHCP

6.2 Static Devices

Device	VLAN	IP Address	Subnet Mask	Gateway	Type
Printer-FAC	220	23.12.145.93	255.255.255.224	23.12.145.65	Static
Printer-MECH	250	23.12.145.189	255.255.255.224	23.12.145.161	Static
SW-154-CORE	240	23.12.145.154	255.255.255.248	23.12.145.153	Static

6.3 Inter-Campus Links

EDGE-99 Gi0/2: 23.12.145.113 | R-154 Gi0/0: 23.12.145.114 | Network: 23.12.145.112/30

6.4 Testing

Same VLAN Ping

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 23.12.145.67

Pinging 23.12.145.67 with 32 bytes of data:

Reply from 23.12.145.67: bytes=32 time=1ms TTL=128
Reply from 23.12.145.67: bytes=32 time=1ms TTL=128
Reply from 23.12.145.67: bytes=32 time<1ms TTL=128
Reply from 23.12.145.67: bytes=32 time<1ms TTL=128

Ping statistics for 23.12.145.67:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Inter VLAN Ping

```
C:\>ping 23.12.145.242

Pinging 23.12.145.242 with 32 bytes of data:

Reply from 23.12.145.242: bytes=32 time<1ms TTL=127
Reply from 23.12.145.242: bytes=32 time<1ms TTL=127
Reply from 23.12.145.242: bytes=32 time<1ms TTL=127
Reply from 23.12.145.242: bytes=32 time<1ms TTL=127

Ping statistics for 23.12.145.242:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Cross Campus Ping

```
C:\>ping 23.12.145.2

Pinging 23.12.145.2 with 32 bytes of data:

Reply from 23.12.145.2: bytes=32 time<1ms TTL=126
Reply from 23.12.145.2: bytes=32 time=1ms TTL=126
Reply from 23.12.145.2: bytes=32 time=12ms TTL=126
Reply from 23.12.145.2: bytes=32 time=12ms TTL=126

Ping statistics for 23.12.145.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 12ms, Average = 6ms
```

100 Campus Cs Lab Pc Ping

```
C:\>ping 23.12.151.2

Pinging 23.12.151.2 with 32 bytes of data:

Reply from 23.12.151.2: bytes=32 time<1ms TTL=125
Reply from 23.12.151.2: bytes=32 time<1ms TTL=125
Reply from 23.12.151.2: bytes=32 time<1ms TTL=125
Reply from 23.12.151.2: bytes=32 time<1ms TTL=125

Ping statistics for 23.12.151.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

ISP Ping

```
C:\>ping 11.11.11.2

Pinging 11.11.11.2 with 32 bytes of data:

Reply from 11.11.11.2: bytes=32 time=2ms TTL=253
Reply from 11.11.11.2: bytes=32 time=2ms TTL=253
Reply from 11.11.11.2: bytes=32 time=2ms TTL=253
Reply from 11.11.11.2: bytes=32 time=2ms TTL=253

Ping statistics for 11.11.11.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 2ms, Average = 2ms
```

Internet Ping

```
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=1ms TTL=253
Reply from 8.8.8.8: bytes=32 time=1ms TTL=253
Reply from 8.8.8.8: bytes=32 time=1ms TTL=253
Reply from 8.8.8.8: bytes=32 time=1ms TTL=253

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

R-154 Router:

R-154

Physical Config CLI Attributes

```
R-154#sh running-config
Building configuration...

Current configuration : 2564 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R-154
!
!
!
enable secret 5 $1$mERr$5.a6P4JqbNiMX01usIfka/
!
!
ip dhcp excluded-address 23.12.145.186
ip dhcp excluded-address 23.12.145.194
ip dhcp excluded-address 23.12.145.129
ip dhcp excluded-address 23.12.145.65
ip dhcp excluded-address 23.12.145.93
ip dhcp excluded-address 23.12.145.94
ip dhcp excluded-address 23.12.145.145
ip dhcp excluded-address 23.12.145.153
ip dhcp excluded-address 23.12.145.154
ip dhcp excluded-address 23.12.145.161
ip dhcp excluded-address 23.12.145.189
```

```
!
ip dhcp pool VLAN210-CLASS154
 network 23.12.145.128 255.255.255.240
 default-router 23.12.145.129
ip dhcp pool VLAN220-FAC154
 network 23.12.145.64 255.255.255.224
 default-router 23.12.145.65
ip dhcp pool VLAN230-ACAD154
 network 23.12.145.144 255.255.255.248
 default-router 23.12.145.145
ip dhcp pool VLAN240-ADMIN154
 network 23.12.145.152 255.255.255.248
 default-router 23.12.145.153
ip dhcp pool VLAN250-MECH154
 network 23.12.145.160 255.255.255.224
 default-router 23.12.145.161
!
!
!
ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO2911/K9 sn FTX152404XT-
!
!
 --More--
```

```
interface GigabitEthernet0/0
 ip address 23.12.145.114 255.255.255.252
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/1.210
 encapsulation dot1Q 210
 ip address 23.12.145.129 255.255.255.240
!
interface GigabitEthernet0/1.220
 encapsulation dot1Q 220
 ip address 23.12.145.65 255.255.255.224
!
interface GigabitEthernet0/1.230
 encapsulation dot1Q 230
 ip address 23.12.145.145 255.255.255.248
!
interface GigabitEthernet0/1.240
 encapsulation dot1Q 240
 ip address 23.12.145.153 255.255.255.248
!
interface GigabitEthernet0/1.250
 encapsulation dot1Q 250
```

```
 ip address 23.12.145.161 255.255.255.224
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
router ospf 1
 log-adjacency-changes
 network 23.12.145.112 0.0.0.3 area 0
 network 23.12.145.128 0.0.0.63 area 0
 network 23.12.145.192 0.0.0.31 area 0
 network 23.12.145.224 0.0.0.7 area 0
 network 23.12.145.232 0.0.0.3 area 0
 network 23.12.145.240 0.0.0.7 area 0
 network 23.12.145.64 0.0.0.31 area 0
!
ip classless
ip route 0.0.0.0 0.0.0.0 23.12.145.113
!
ip flow-export version 9
!
```

SW-154-CORE:

SW-154-CORE

Physical Config CLI Attributes

```
SW-154-CORE#sh running-config
Building configuration...

Current configuration : 1915 bytes
!
version 16.3.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname SW-154-CORE
!
!
no profinet
enable secret 5 $1$mERr$5.a6P4JqbNiMX01usIfka/
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
```

```
interface GigabitEthernet1/0/1
 switchport trunk allowed vlan 210,220,230,240,250
 switchport mode trunk
!
interface GigabitEthernet1/0/2
 switchport trunk allowed vlan 220
 switchport mode trunk
 spanning-tree portfast trunk
!
interface GigabitEthernet1/0/3
 switchport trunk allowed vlan 210
 switchport mode trunk
 spanning-tree portfast trunk
!
interface GigabitEthernet1/0/4
 switchport trunk allowed vlan 250
 switchport mode trunk
 spanning-tree portfast trunk
!
interface GigabitEthernet1/0/5
!
interface GigabitEthernet1/0/6
!
interface GigabitEthernet1/0/7
!
interface GigabitEthernet1/0/8
!
interface GigabitEthernet1/0/9
 --More--
```

```
interface GigabitEthernet1/0/10
!
interface GigabitEthernet1/0/11
!
interface GigabitEthernet1/0/12
!
interface GigabitEthernet1/0/13
!
interface GigabitEthernet1/0/14
!
interface GigabitEthernet1/0/15
!
interface GigabitEthernet1/0/16
!
interface GigabitEthernet1/0/17
!
interface GigabitEthernet1/0/18
!
interface GigabitEthernet1/0/19
!
interface GigabitEthernet1/0/20
!
interface GigabitEthernet1/0/21
!
interface GigabitEthernet1/0/22
!
interface GigabitEthernet1/0/23
!
 --More--
```

```
interface GigabitEthernet1/0/24
!
interface GigabitEthernet1/1/1
!
interface GigabitEthernet1/1/2
!
interface GigabitEthernet1/1/3
!
interface GigabitEthernet1/1/4
!
interface Vlan1
 no ip address
 shutdown
!
interface Vlan240
 mac-address 0005.5e47.de01
 ip address 23.12.145.154 255.255.255.248
!
ip default-gateway 23.12.145.153
ip classless
!
ip flow-export version 9
!
```

7. Campus 153

Campus 153 connects to EDGE-99 via a Serial link. It is a smaller campus with 3 VLANs serving classrooms, admission, and academic departments.

7.1 VLAN Configuration

VLAN	Name	Network	Gateway	DHCP Range	Type
260	CLASSROOMS	192.168.3.16/28	192.168.3.17	.18–.30	DHCP
270	ADMISSION	192.168.3.0/28	192.168.3.1	.2–.14	DHCP
280	ACADEMIC	192.168.3.32/29	192.168.3.33	.34–.38	DHCP

7.2 Static Devices

Device	VLAN	IP Address	Subnet Mask	Gateway	Type
Printer-Admission	270	192.168.3.14	255.255.255.240	192.168.3.1	Static
Printer-Academic	280	192.168.3.38	255.255.255.248	192.168.3.33	Static

7.3 Devices Summary

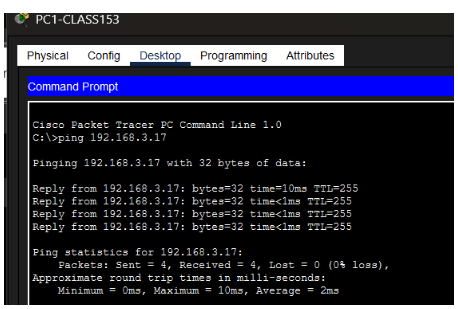
- PC1-Class (VLAN 260): 192.168.3.18 — DHCP
- PC2-Class (VLAN 260): 192.168.3.19 — DHCP
- PC1-Admission (VLAN 270): 192.168.3.2 — DHCP
- PC2-Admission (VLAN 270): 192.168.3.3 — DHCP
- PC-Academic (VLAN 280): 192.168.3.34 — DHCP

7.4 Serial Link

EDGE-99 Se0/0/0: 23.12.145.117 | R-153 Se0/0/0: 23.12.145.118 | Network: 23.12.145.116/30

7.5 Testing

Same VLAN Ping



Inter-VLAN Ping

```
C:\>ping 192.168.3.1

Pinging 192.168.3.1 with 32 bytes of data:

Reply from 192.168.3.1: bytes=32 time<1ms TTL=255
Reply from 192.168.3.1: bytes=32 time<1ms TTL=255
Reply from 192.168.3.1: bytes=32 time<1ms TTL=255
Reply from 192.168.3.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.3.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Cross-Campus Ping

```
C:\>ping 23.12.145.2

Pinging 23.12.145.2 with 32 bytes of data:

Request timed out.
Reply from 23.12.145.2: bytes=32 time=11ms TTL=126
Reply from 23.12.145.2: bytes=32 time=2ms TTL=126
Reply from 23.12.145.2: bytes=32 time=4ms TTL=126

Ping statistics for 23.12.145.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 11ms, Average = 5ms
```

ISP Ping

```
C:\>ping 11.11.11.2

Pinging 11.11.11.2 with 32 bytes of data:

Reply from 11.11.11.2: bytes=32 time=4ms TTL=253
Reply from 11.11.11.2: bytes=32 time=4ms TTL=253
Reply from 11.11.11.2: bytes=32 time=4ms TTL=253
Reply from 11.11.11.2: bytes=32 time=4ms TTL=253

Ping statistics for 11.11.11.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 4ms, Average = 4ms

C:\>
```

Internet Ping

```
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=4ms TTL=253
Reply from 8.8.8.8: bytes=32 time=3ms TTL=253
Reply from 8.8.8.8: bytes=32 time=3ms TTL=253
Reply from 8.8.8.8: bytes=32 time=2ms TTL=253

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 4ms, Average = 3ms
```

R-153 Router:

R-153

Physical Config CLI Attributes

```
R-153#sh running-config
Building configuration...

Current configuration : 1960 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R-153
!
!
!
enable secret 5 $1$mERr$5.a6P4JqbNiMX01usIfka/
!
!
ip dhcp excluded-address 192.168.3.1
ip dhcp excluded-address 192.168.3.17
ip dhcp excluded-address 192.168.3.33
ip dhcp excluded-address 192.168.3.29
!
ip dhcp pool ADMISSION
 network 192.168.3.0 255.255.255.240
 default-router 192.168.3.1
 dns-server 8.8.8.8
ip dhcp pool CLASSROOMS
 network 192.168.3.16 255.255.255.240
```

```
 default-router 192.168.3.17
 dns-server 8.8.8.8
ip dhcp pool ACADEMIC
 network 192.168.3.32 255.255.255.248
 default-router 192.168.3.33
 dns-server 8.8.8.8
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO2911/K9 sn FTX1524TUSB-
!
```

```
interface GigabitEthernet0/0
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/0.260
 encapsulation dot1Q 260
 ip address 192.168.3.17 255.255.255.240
!
interface GigabitEthernet0/0.270
 encapsulation dot1Q 270
 ip address 192.168.3.1 255.255.255.240
!
interface GigabitEthernet0/0.280
 encapsulation dot1Q 280
 ip address 192.168.3.33 255.255.255.248
!
interface GigabitEthernet0/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
```

```
interface Serial0/0/0
 ip address 23.12.145.118 255.255.255.252
 clock rate 64000
!
interface Serial0/0/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
router ospf 1
 router-id 2.2.2.2
 log-adjacency-changes
 network 192.168.3.0 0.0.0.255 area 0
 network 11.11.11.0 0.0.0.3 area 0
 network 23.12.145.0 0.0.0.255 area 0
 network 23.12.151.0 0.0.0.255 area 0
!
ip classless
ip route 0.0.0.0 0.0.0.0 23.12.145.117
!
ip flow-export version 9
!
```

SW-153-CORE:

SW-153-CORE

Physical Config CLI Attributes

```
SW-153-CORE#sh running-config
Building configuration...

Current configuration : 1753 bytes
!
version 16.3.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname SW-153-CORE
!
!
no profinet
enable secret 5 $1$mERr$5.a6P4JqbNiMX01usIfka/
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
```

```
interface GigabitEthernet1/0/1
 switchport trunk allowed vlan 260,270,280
 switchport mode trunk
!
interface GigabitEthernet1/0/2
 switchport trunk allowed vlan 260,270,280
 switchport mode trunk
!
interface GigabitEthernet1/0/3
!
interface GigabitEthernet1/0/4
!
interface GigabitEthernet1/0/5
!
interface GigabitEthernet1/0/6
!
interface GigabitEthernet1/0/7
!
interface GigabitEthernet1/0/8
!
interface GigabitEthernet1/0/9
!
interface GigabitEthernet1/0/10
!
interface GigabitEthernet1/0/11
!
interface GigabitEthernet1/0/12
```

```
!
interface GigabitEthernet1/0/13
!
interface GigabitEthernet1/0/14
!
interface GigabitEthernet1/0/15
!
interface GigabitEthernet1/0/16
!
interface GigabitEthernet1/0/17
!
interface GigabitEthernet1/0/18
!
interface GigabitEthernet1/0/19
!
interface GigabitEthernet1/0/20
!
interface GigabitEthernet1/0/21
!
interface GigabitEthernet1/0/22
!
interface GigabitEthernet1/0/23
!
interface GigabitEthernet1/0/24
!
interface GigabitEthernet1/1/1
!
interface GigabitEthernet1/1/2
```

```
!
interface GigabitEthernet1/1/3
!
interface GigabitEthernet1/1/4
!
interface Vlan1
 no ip address
 shutdown
!
interface Vlan260
 mac-address 0060.3e7d.7001
 ip address 192.168.3.29 255.255.255.240
!
ip default-gateway 192.168.3.17
ip classless
!
ip flow-export version 9
!
!
```

8. Campus 79 — Administrative Campus

Campus 79 is an administrative campus hosting 6 VLANs for departments such as HR, Finance, Procurement, Admin, Library, and EDC. The topology was reduced to 80% of original devices for simulation.

8.1 VLAN Configuration

VLAN	Name	Network	Gateway	Usable Range	Type
310	LIBRARY-79	192.168.4.0/28	192.168.4.1	.2–.14	DHCP
320	HR-79	192.168.4.16/28	192.168.4.17	.18–.30	DHCP
330	PROCUREMENT-79	192.168.4.32/28	192.168.4.33	.34–.46	DHCP
340	ADMIN-79	192.168.4.48/28	192.168.4.49	.50–.62	DHCP
350	FINANCE-79	192.168.4.64/28	192.168.4.65	.66–.78	DHCP
360	EDC-79	192.168.4.80/28	192.168.4.81	.82–.94	DHCP

8.2 Static Devices

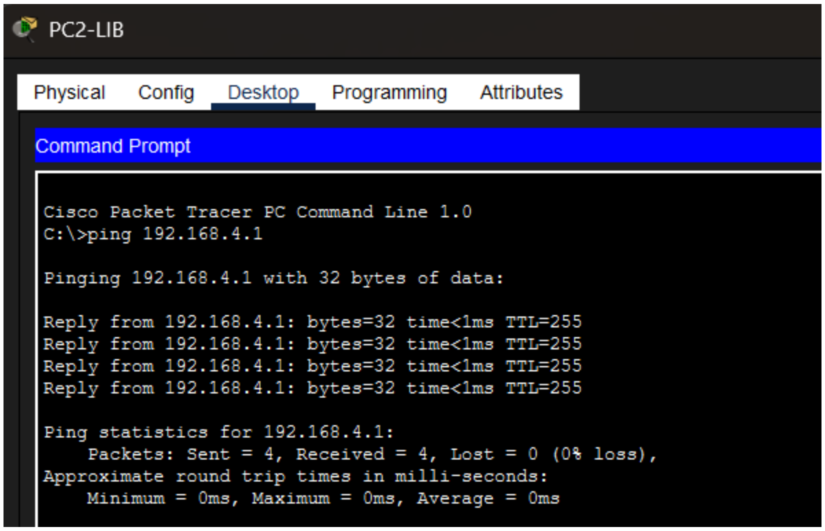
Device	VLAN	IP Address	Subnet Mask	Gateway	Type
Printer-HR-1	320	192.168.4.29	255.255.255.240	192.168.4.17	Static
Printer-HR-2	320	192.168.4.30	255.255.255.240	192.168.4.17	Static
Printer-PROC	330	192.168.4.45	255.255.255.240	192.168.4.33	Static
Printer-ADMIN-1	340	192.168.4.61	255.255.255.240	192.168.4.49	Static
Printer-ADMIN-2	340	192.168.4.62	255.255.255.240	192.168.4.49	Static
Printer-FIN-1	350	192.168.4.77	255.255.255.240	192.168.4.65	Static
Printer-FIN-2	350	192.168.4.78	255.255.255.240	192.168.4.65	Static
Printer-EDC	360	192.168.4.93	255.255.255.240	192.168.4.81	Static
SW-79-CORE	340	192.168.4.50	255.255.255.240	192.168.4.49	Static

8.3 Reduced Topology Note

As per project requirements, device count was reduced by 80% from the original specification. Each department retains 1 PC and 1 Printer (where applicable) for simulation purposes.

8.4 Testing

Same VLAN Ping



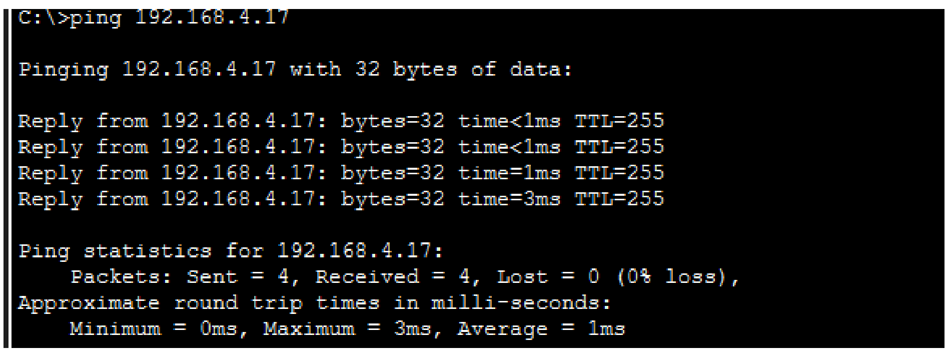

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.4.1

Pinging 192.168.4.1 with 32 bytes of data:

Reply from 192.168.4.1: bytes=32 time<1ms TTL=255
Reply from 192.168.4.1: bytes=32 time<1ms TTL=255
Reply from 192.168.4.1: bytes=32 time<1ms TTL=255
Reply from 192.168.4.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.4.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Inter-VLAN Ping



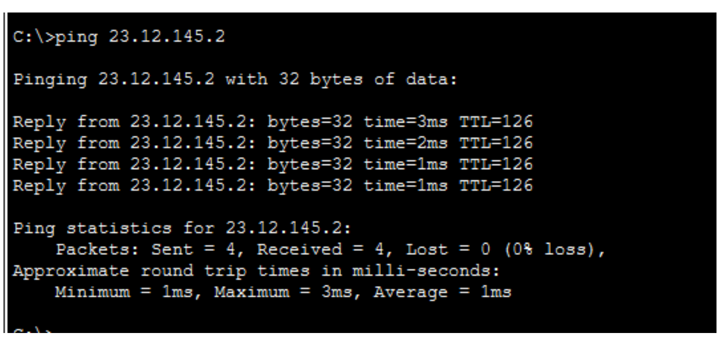
```
C:\>ping 192.168.4.17

Pinging 192.168.4.17 with 32 bytes of data:

Reply from 192.168.4.17: bytes=32 time<1ms TTL=255
Reply from 192.168.4.17: bytes=32 time<1ms TTL=255
Reply from 192.168.4.17: bytes=32 time=1ms TTL=255
Reply from 192.168.4.17: bytes=32 time=3ms TTL=255

Ping statistics for 192.168.4.17:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 1ms
```

Cross-Campus Ping



```
C:\>ping 23.12.145.2

Pinging 23.12.145.2 with 32 bytes of data:

Reply from 23.12.145.2: bytes=32 time=3ms TTL=126
Reply from 23.12.145.2: bytes=32 time=2ms TTL=126
Reply from 23.12.145.2: bytes=32 time=1ms TTL=126
Reply from 23.12.145.2: bytes=32 time=1ms TTL=126

Ping statistics for 23.12.145.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 1ms
```

ISP Ping

```
C:\>ping 11.11.11.2

Pinging 11.11.11.2 with 32 bytes of data:

Reply from 11.11.11.2: bytes=32 time=4ms TTL=253
Reply from 11.11.11.2: bytes=32 time=4ms TTL=253
Reply from 11.11.11.2: bytes=32 time=4ms TTL=253
Reply from 11.11.11.2: bytes=32 time=4ms TTL=253

Ping statistics for 11.11.11.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 4ms, Average = 4ms

C:\>
```

Internet Ping

```
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=4ms TTL=253
Reply from 8.8.8.8: bytes=32 time=3ms TTL=253
Reply from 8.8.8.8: bytes=32 time=3ms TTL=253
Reply from 8.8.8.8: bytes=32 time=2ms TTL=253

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 4ms, Average = 3ms
```

R-79 Router:

R-79

Physical Config CLI Attributes

```
R-79#sh running-config
Building configuration...

Current configuration : 3021 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname R-79
!
!
!
enable secret 5 $1$mERr$5.a6P4JqbNiMX01usIfka/
!
!
ip dhcp excluded-address 192.168.4.1
ip dhcp excluded-address 192.168.4.17
ip dhcp excluded-address 192.168.4.29
ip dhcp excluded-address 192.168.4.30
ip dhcp excluded-address 192.168.4.33
ip dhcp excluded-address 192.168.4.45
ip dhcp excluded-address 192.168.4.49
ip dhcp excluded-address 192.168.4.50
ip dhcp excluded-address 192.168.4.61
ip dhcp excluded-address 192.168.4.62
ip dhcp excluded-address 192.168.4.65
ip dhcp excluded-address 192.168.4.77
ip dhcp excluded-address 192.168.4.78
```

```
ip dhcp excluded-address 192.168.4.81
ip dhcp excluded-address 192.168.4.93
!
ip dhcp pool VLAN310-LIB79
 network 192.168.4.0 255.255.255.240
 default-router 192.168.4.1
ip dhcp pool VLAN320-HR79
 network 192.168.4.16 255.255.255.240
 default-router 192.168.4.17
ip dhcp pool VLAN330-PROC79
 network 192.168.4.32 255.255.255.240
 default-router 192.168.4.33
ip dhcp pool VLAN340-ADMIN79
 network 192.168.4.48 255.255.255.240
 default-router 192.168.4.49
ip dhcp pool VLAN350-FIN79
 network 192.168.4.64 255.255.255.240
 default-router 192.168.4.65
ip dhcp pool VLAN360-EDC79
 network 192.168.4.80 255.255.255.240
 default-router 192.168.4.81
!
!
!
no ip cef
no ipv6 cef

.
interface GigabitEthernet0/0
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/0.310
 encapsulation dot1Q 310
 ip address 192.168.4.1 255.255.255.240
!
interface GigabitEthernet0/0.320
 encapsulation dot1Q 320
 ip address 192.168.4.17 255.255.255.240
!
interface GigabitEthernet0/0.330
 encapsulation dot1Q 330
 ip address 192.168.4.33 255.255.255.240
!
interface GigabitEthernet0/0.340
 encapsulation dot1Q 340
 ip address 192.168.4.49 255.255.255.240
!
interface GigabitEthernet0/0.350
 encapsulation dot1Q 350
 ip address 192.168.4.65 255.255.255.240
!
interface GigabitEthernet0/0.360
 encapsulation dot1Q 360
 ip address 192.168.4.81 255.255.255.240
!
interface GigabitEthernet0/1
 --More--
```

```
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Serial0/0/0
 ip address 23.12.145.122 255.255.255.252
!
interface Serial0/0/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
router ospf 1
 log-adjacency-changes
 network 23.12.145.120 0.0.0.3 area 0
 network 192.168.4.0 0.0.0.15 area 0
```

```
 network 192.168.4.16 0.0.0.15 area 0
 network 192.168.4.32 0.0.0.15 area 0
 network 192.168.4.48 0.0.0.15 area 0
 network 192.168.4.64 0.0.0.15 area 0
 network 192.168.4.80 0.0.0.15 area 0
!
ip classless
!
ip flow-export version 9
!
!
!
banner motd ^CUnauthorized Access Prohibited^C
!
!
!
!
```

SW-79-CORE:

SW-79-CORE

Physical Config CLI Attributes

```
SW-79-CORE#sh running-config
Building configuration...

Current configuration : 2021 bytes
!
version 16.3.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname SW-79-CORE
!
!
no profinet
enable secret 5 $1$mERr$5.a6P4JqbNiMX01usIfka/
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
```

```
!
interface GigabitEthernet1/0/1
 switchport trunk allowed vlan 310,320,330,340,350,360
 switchport mode trunk
!
interface GigabitEthernet1/0/2
 switchport trunk allowed vlan 310
 switchport mode trunk
!
interface GigabitEthernet1/0/3
 switchport trunk allowed vlan 320
 switchport mode trunk
!
interface GigabitEthernet1/0/4
 switchport trunk allowed vlan 330
 switchport mode trunk
!
interface GigabitEthernet1/0/5
 switchport trunk allowed vlan 340
 switchport mode trunk
!
interface GigabitEthernet1/0/6
 switchport trunk allowed vlan 350
 switchport mode trunk
!
interface GigabitEthernet1/0/7
 switchport trunk allowed vlan 360
 switchport mode trunk
!
```

```
interface GigabitEthernet1/0/8
!
interface GigabitEthernet1/0/9
!
interface GigabitEthernet1/0/10
!
interface GigabitEthernet1/0/11
!
interface GigabitEthernet1/0/12
!
interface GigabitEthernet1/0/13
!
interface GigabitEthernet1/0/14
!
interface GigabitEthernet1/0/15
!
interface GigabitEthernet1/0/16
!
interface GigabitEthernet1/0/17
!
interface GigabitEthernet1/0/18
!
interface GigabitEthernet1/0/19
!
interface GigabitEthernet1/0/20
!
interface GigabitEthernet1/0/21
!
```

```
!
interface GigabitEthernet1/0/22
!
interface GigabitEthernet1/0/23
!
interface GigabitEthernet1/0/24
!
interface GigabitEthernet1/1/1
!
interface GigabitEthernet1/1/2
!
interface GigabitEthernet1/1/3
!
interface GigabitEthernet1/1/4
!
interface Vlan1
 no ip address
 shutdown
!
interface Vlan340
 mac-address 00d0.d381.e501
 ip address 192.168.4.50 255.255.255.240
!
ip default-gateway 192.168.4.49
ip classless
!
ip flow-export version 9
!
```

9. Media Villa

Media Villa is connected to EDGE-99 via a Serial link. It hosts 3 VLANs for classrooms, faculty, and academic departments. Device count was also reduced for simulation.

9.1 VLAN Configuration

VLAN	Name	Network	Gateway	DHCP Range	Type
380	CLASSROOMS-MEDIA	192.168.5.0/29	192.168.5.1	.2–.6	DHCP
390	FACULTY-MEDIA	192.168.5.16/28	192.168.5.17	.18–.26	DHCP
400	ACADEMIC-MEDIA	192.168.5.32/29	192.168.5.33	.34–.38	DHCP

9.2 Static Devices

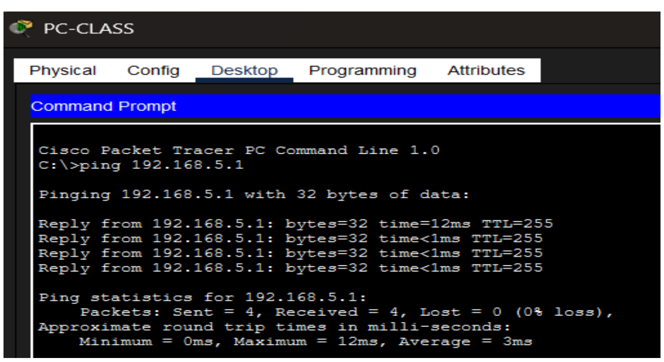
Device	VLAN	IP Address	Subnet Mask	Gateway	Type
Printer-FAC	390	192.168.5.27	255.255.255.240	192.168.5.17	Static
Printer-ACAD	400	192.168.5.38	255.255.255.248	192.168.5.33	Static
SW-MEDIA-CORE	380	192.168.5.3	255.255.255.248	192.168.5.1	Static

9.3 Serial Link

EDGE-99 Se0/1/1: 23.12.145.125 | R-MEDIA Se0/1/0: 23.12.145.126 | Network: 23.12.145.124/30

9.4 Testing

Same VLAN Ping



Inter-VLAN Ping

```
C:\>ping 192.168.5.17

Pinging 192.168.5.17 with 32 bytes of data:

Reply from 192.168.5.17: bytes=32 time<1ms TTL=255
Reply from 192.168.5.17: bytes=32 time<1ms TTL=255
Reply from 192.168.5.17: bytes=32 time=10ms TTL=255
Reply from 192.168.5.17: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.5.17:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms
```

Cross-Campus Ping

```
C:\>ping 192.168.3.14

Pinging 192.168.3.14 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.14: bytes=32 time=3ms TTL=125
Reply from 192.168.3.14: bytes=32 time=2ms TTL=125
Reply from 192.168.3.14: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.3.14:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 3ms, Average = 2ms
```

ISP Ping

```
C:\>ping 11.11.11.2

Pinging 11.11.11.2 with 32 bytes of data:

Reply from 11.11.11.2: bytes=32 time=4ms TTL=253
Reply from 11.11.11.2: bytes=32 time=4ms TTL=253
Reply from 11.11.11.2: bytes=32 time=4ms TTL=253
Reply from 11.11.11.2: bytes=32 time=4ms TTL=253

Ping statistics for 11.11.11.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 4ms, Average = 4ms

C:\>
```

Internet Ping

```
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=4ms TTL=253
Reply from 8.8.8.8: bytes=32 time=3ms TTL=253
Reply from 8.8.8.8: bytes=32 time=3ms TTL=253
Reply from 8.8.8.8: bytes=32 time=2ms TTL=253

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 4ms, Average = 3ms
```

R-MEDIA Router:

```
R-MEDIA>en
Password:
R-MEDIA#sh running-config
Building configuration...

Current configuration : 2331 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R-MEDIA
!
!
!
enable secret 5 $1$mERr$5.a6P4JqbNiMX01usIfka/
!
!
ip dhcp excluded-address 192.168.5.1
ip dhcp excluded-address 192.168.5.17
ip dhcp excluded-address 192.168.5.27
ip dhcp excluded-address 192.168.5.33
ip dhcp excluded-address 192.168.5.38
!
```

```
!
 --More--
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to down

04:45:29: %OSPF-5-ADJCHG: Process 1, Nbr 23.12.145.125 on Serial0/1/0 from FULL to DOWN, Neighbor Down:
Interface down or detached

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up
ip dhcp pool VLAN380-CLASS-MEDIA
 network 192.168.5.0 255.255.255.248
 default-router 192.168.5.1
 dns-server 8.8.8.8
ip dhcp pool VLAN390-FAC-MEDIA
 network 192.168.5.16 255.255.255.240
 default-router 192.168.5.17
 dns-server 8.8.8.8
ip dhcp pool VLAN400-ACAD-MEDIA
 network 192.168.5.32 255.255.255.248
 default-router 192.168.5.33
 dns-server 8.8.8.8
!
!
!
no ip cef
no ipv6 cef
!
!
```

```
!
interface GigabitEthernet0/0
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/0.380
 encapsulation dot1Q 380
 ip address 192.168.5.1 255.255.255.248
!
interface GigabitEthernet0/0.390
 encapsulation dot1Q 390
 ip address 192.168.5.17 255.255.255.240
!
interface GigabitEthernet0/0.400
 encapsulation dot1Q 400
 ip address 192.168.5.33 255.255.255.248
!
interface GigabitEthernet0/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
```

```
 shutdown
!
interface Serial0/0/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/0/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/1/0
 ip address 23.12.145.126 255.255.255.252
!
interface Serial0/1/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/2/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/2/1
 no ip address
 clock rate 2000000
```

```
 shutdown
!
interface Serial0/3/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/3/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
router ospf 1
 log-adjacency-changes
 network 23.12.145.124 0.0.0.3 area 0
 network 192.168.5.0 0.0.0.255 area 0
!
ip classless
ip route 0.0.0.0 0.0.0.0 23.12.145.125
!
ip flow-export version 9
!
!
```

SW-MEDIA-CORE:

```
SW-MEDIA-CORE#sh running-config
Building configuration...

Current configuration : 1343 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname SW-MEDIA-CORE
!
enable secret 5 $1$mERr$5.a6P4JqbNiMX01usIfka/
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
```

```
 switchport mode trunk
!
interface FastEthernet0/2
 switchport mode trunk
!
interface FastEthernet0/3
 switchport mode trunk
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
```

```
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
```

```
 no ip address
 shutdown
!
interface Vlan380
 ip address 192.168.5.3 255.255.255.248
!
ip default-gateway 192.168.5.1
!
!
!
```

10. IP Address Summary

10.1 Address Blocks Used

Block	Campus / Usage	Notes
23.12.145.0/24	Campus 99 (EDGE-99) VLANs + WAN links	Shared across C99, C154, WAN subnets
23.12.151.0/24	Campus 100 — Labs 10–40 (CS, AI, L3, L4)	Public-class addressing
192.168.1.0/24	Campus 100 — Labs 50–70	Private RFC1918
192.168.2.0/24	Campus 100 — Remaining VLANs	Private RFC1918
192.168.3.0/24	Campus 153	Private RFC1918
192.168.4.0/24	Campus 79	Private RFC1918
192.168.5.0/24	Media Villa	Private RFC1918
11.11.11.0/30	ISP WAN Link	EDGE-99 ↔ ISP

10.2 DHCP vs Static Summary

Campus	DHCP Pools	Static Devices
Campus 99	7 pools	6 (4 servers + 2 printers)
Campus 100	13 pools	13 (FTP servers, printers, core switch)
Campus 154	5 pools	3 (2 printers + core switch)
Campus 153	3 pools	2 printers
Campus 79	6 pools	9 (printers + core switch)
Media Villa	3 pools	3 (2 printers + core switch)

11. Design Decisions & Justifications

11.1 VLAN Segmentation

Each department is assigned a dedicated VLAN to isolate broadcast domains, improve security, and simplify management. This prevents unauthorized cross-department access at Layer 2.

11.2 Router-on-a-Stick (Sub-interfaces)

Inter-VLAN routing is implemented using sub-interfaces on each campus router. This approach minimizes hardware requirements while enabling full Layer 3 communication between VLANs.

11.3 DHCP for Dynamic Hosts

All end-user devices (PCs) are configured with DHCP to simplify administration and reduce manual configuration errors. Each VLAN has a dedicated DHCP pool on the campus router.

11.4 Static IPs for Critical Infrastructure

Servers, printers, and core switches are assigned static IP addresses to ensure consistent reachability, facilitate DNS resolution, and simplify network monitoring.

11.5 Centralized Hub via EDGE-99

EDGE-99 serves as the single point of internet access and inter-campus routing hub. This simplifies routing table management and provides a single point for security policy enforcement.

11.6 Subnet Sizing

Subnets were sized using VLSM (Variable Length Subnet Masking) to match the number of hosts per department, minimizing IP address waste while providing room for growth.

12. Conclusion

This project successfully demonstrates the design and configuration of a scalable, segmented multi-campus enterprise network. The network supports 6 campuses with 30+ VLANs, centralized internet access, DHCP-based dynamic addressing for users, and static addressing for critical infrastructure.

The hub-and-spoke topology centered on EDGE-99 provides simplified management and routing while ensuring all campuses remain interconnected. The use of VLSM ensures efficient utilization of the assigned IP address space.

The simulation in Cisco Packet Tracer validates the design, confirming successful inter-VLAN communication, DHCP assignment, and end-to-end connectivity across all campuses.

13. References

- Cisco Systems — Cisco Packet Tracer 8.x Documentation
- Tanenbaum, A. S. & Wetherall, D. — Computer Networks, 5th Edition
- RFC 1918 — Address Allocation for Private Internets
- Cisco Press — CCNA Routing and Switching 200-125 Official Cert Guide
- Course Materials — CNDC Lecture Slides & Lab Manuals